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United States Patent Application Publication	20250270684
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	FUKAYA; YOUICHI

VAPOR DEPOSITION MASK AND METHOD FOR MANUFACTURING ORGANIC ELECTRONIC DEVICE

Abstract

A vapor deposition mask includes a base member made of a non-magnetic material. The base member includes a first region having a plurality of through holes and a second region thicker than the first region. The second region has a recess including a magnetic layer therein.

Inventors:	**FUKAYA; YOUICHI (Tokyo, JP)**
Applicant:	**CANON KABUSHIKI KAISHA** (Tokyo, JP)
Family ID:	**1000008493313**
Appl. No.:	**19/063033**
Filed:	**February 25, 2025**

Foreign Application Priority Data

JP	2024-028534	Feb. 28, 2024

Publication Classification

Int. Cl.: **C23C14/04** (20060101); **H10K71/16** (20230101)

U.S. Cl.:

CPC **C23C14/042** (20130101); **H10K71/166** (20230201);

Background/Summary

BACKGROUND
Technical Field

[0001] The present disclosure relates to a vapor deposition mask and a method for manufacturing an organic electronic device.

Description of the Related Art

[0002] Organic electroluminescence (EL) elements are light-emitting display elements having a layered structure of thin films and capable of responding at high speed. Organic EL panels are lightweight and can constitute display devices highly capable of displaying moving images, and therefore have attracted considerable attention as display devices, such as flat-panel display (FPD) televisions or small displays for electronic viewfinders (EVFs).

[0003] Organic EL display devices are often manufactured by vapor-depositing an organic material using a vacuum vapor deposition apparatus of resistance heating type. Particularly in full-color organic EL display devices, fine red, green and blue (RGB) light-emitting elements need to be produced with high accuracy. The full-color organic EL display devices are thus manufactured with a mask vapor deposition technique in which a vapor deposition mask made of a magnetic metal or the like is used to selectively vacuum-deposit different organic materials for respective RGB pixels at desired positions. The mask vapor deposition technique is a technique with which a vivid full-color organic EL display device is manufactured by vapor-depositing materials through a vapor deposition mask that is closely attracted by a magnet or the like to a substrate to be subjected to vapor deposition.

[0004] Vapor deposition masks are often made of a magnetic material. This is because a vapor deposition mask made of a magnetic material can be closely attached, by using a magnet or the like, relatively easily to a substrate to be subjected to vapor deposition. However, such a vapor deposition mask made of a magnetic material has some drawbacks which include its heavy weight, difficulty in processing with high accuracy, and susceptibility to plastic deformation.

[0005] To reduce the weight of a vapor deposition mask, Japanese Patent Laid-Open No. 2017-20068 discloses a technique in which a resin mask has recesses or through holes that are provided with a magnetic member therein. With this technique, it is possible to reduce the weight of the mask and improve adhesion between the vapor deposition mask and a substrate to be subjected to vapor deposition.

[0006] Japanese Patent Laid-Open No. 2001-185350 proposes a technique in which a vapor deposition mask is manufactured using a silicon substrate as a base material that can be processed with high accuracy and is resistant to plastic deformation. Since silicon can be processed using a semiconductor manufacturing technique, processing accuracy can be improved to several micrometers.

[0007] Organic EL display devices have become increasingly high-definition in recent years. Thus, since high-precision processing is required, a vapor deposition mask is extremely thin. This creates a challenge in attracting the vapor deposition mask with magnetic force.

[0008] In the vapor deposition mask described in Japanese Patent Laid-Open No. 2017-20068, a thinner resin mask means shallower recesses or through holes and this means, as a result, that a magnetic material applied is also thinner. Thus, the volume of the magnetic material decreases, and it is difficult to obtain a magnetic force that is large enough to bring the vapor deposition mask into contact with a substrate to be subjected to vapor deposition.

[0009] Another challenge created when the vapor deposition mask is thinner is that the vapor deposition mask is prone to plastic deformation or damage. Even if a thinner region of the vapor deposition mask can be reliably attracted by magnetic force, the region made of, for example, resin or metal may be plastically deformed by stress applied thereto. If the thinner region is constituted by a silicon substrate, as described in Japanese Patent Laid-Open No. 2001-185350, the thinner region may be too fragile to withstand the magnetic force and may be damaged.

SUMMARY

[0010] A vapor deposition mask according to an aspect of the present disclosure includes a base member made of a non-magnetic material. The base member includes a first region having a

plurality of through holes and a second region thicker than the first region. The second region has a recess including a magnetic layer therein.
[0011] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1**A and FIG. **1**B schematically illustrate a vapor deposition mask according to a first embodiment.
[0013] FIG. **2** is a schematic cross-sectional view illustrating how vapor deposition is performed using the vapor deposition mask illustrated in FIG. **1**A and FIG. **1**B.
[0014] FIG. **3**A and FIG. **3**B are schematic cross-sectional views each illustrating a comparative example of the present disclosure.
[0015] FIG. **4**A to FIG. **4**D are schematic cross-sectional views illustrating an example of a method for manufacturing the vapor deposition mask according to the first embodiment.
[0016] FIG. **5**A to FIG. **5**C are schematic cross-sectional views illustrating the example of the method for manufacturing the vapor deposition mask according to the first embodiment.
[0017] FIG. **6** is a schematic cross-sectional view illustrating how vapor deposition is performed using a vapor deposition mask according to a second embodiment.
[0018] FIG. **7**A to FIG. **7**D are schematic cross-sectional views illustrating an example of a method for manufacturing the vapor deposition mask according to the second embodiment.
[0019] FIG. **8** is a schematic cross-sectional view illustrating how vapor deposition is performed using a vapor deposition mask according to a third embodiment.
[0020] FIG. **9**A to FIG. **9**D are schematic cross-sectional views illustrating an example of a method for manufacturing the vapor deposition mask according to the third embodiment.
[0021] FIG. **10** schematically illustrates an example of a display device according to an embodiment of the present disclosure.
[0022] FIG. **11**A schematically illustrates an example of an imaging device according to an embodiment of the present disclosure, and FIG. **11**B schematically illustrates an example of an electronic apparatus according to an embodiment of the present disclosure.
[0023] FIG. **12**A schematically illustrates an example of a display device according to an embodiment of the present disclosure, and FIG. **12**B schematically illustrates an example of a foldable display device according to an embodiment of the present disclosure.
[0024] FIG. **13**A schematically illustrates an example of an illuminating apparatus according to an embodiment of the present disclosure, and FIG. **13**B schematically illustrates an example of a moving body including a vehicle lighting tool according to an embodiment of the present disclosure.
[0025] FIG. **14**A schematically illustrates an example of a wearable device according to an embodiment of the present disclosure, and FIG. **14**B schematically illustrates another example of the wearable device according to the embodiment of the present disclosure.
[0026] FIG. **15**A schematically illustrates an example of an image forming apparatus according to an embodiment of the present disclosure, and FIG. **15**B and FIG. **15**C each schematically illustrate an example of an exposing source of the image forming apparatus according to the embodiment of the present disclosure.
DESCRIPTION OF THE EMBODIMENTS
[0027] Embodiments of the present disclosure will now be described with reference to the drawings. Note that the present disclosure can be implemented in many different forms, and is not to be interpreted as being limited to the description of the embodiments set forth herein. For more

clarification of the description, the widths, thicknesses, shapes, and the like of components of actual embodiments may be schematically illustrated in the drawings. However, they are merely examples and are not to limit the interpretation of the present disclosure.

First Embodiment

[0028] A first embodiment of the present disclosure will now be described.

Vapor Deposition Mask

[0029] FIG. **1**A and FIG. **1**B schematically illustrate a vapor deposition mask according to the first embodiment of the present disclosure. FIG. **1**A is a schematic plan view as seen from a surface facing a substrate to be subjected to vapor deposition. FIG. **1**B is a schematic view of a cross-section taken along line IB-IB in FIG. **1**A.

[0030] A vapor deposition mask **100** illustrated in FIG. **1**A and FIG. **1**B includes a base member **1** made of a non-magnetic material. As illustrated in FIG. **1**B, the base member **1** includes a membrane region (first region) **2** having a plurality of openings (through holes) **4** and a beam region (second region) **3** thicker than the membrane region **2**. The membrane region **2** is disposed to be flush with a surface of the beam region **3** having a magnetic layer **6** therein. As illustrated in FIG. **1**A, the base member **1** is circular and includes a plurality of rectangular membrane regions **2**. The beam region **3** is disposed around the membrane regions **2** and at an end portion of the base member **1**. The beam region **3** has a recess **5** in the form of a grid-like groove and also in the form of an annular groove along the outer periphery of the base member **1**. The depth of the recess **5** is greater than the thickness of the membrane regions **2**. The magnetic layer **6** is embedded in the recess **5**. The thickness of the magnetic layer **6** is greater than the depth of the recess **5**. A low-hardness layer **7** is disposed on the surface of the magnetic layer **6**.

[0031] FIG. **2** is a schematic cross-sectional view illustrating how vapor deposition is performed using the vapor deposition mask illustrated in FIG. **1**A and FIG. **1**B. As illustrated in FIG. **2**, a substrate to be subjected to vapor deposition (hereinafter simply referred to as substrate) **8** is held by an electrostatic chuck **9**, with a surface to be subjected to vapor deposition facing downward. A magnet **10** is disposed on a surface of the electrostatic chuck **9** opposite a surface thereof adjacent to the substrate **8**. The vapor deposition mask **100** is disposed such that a surface thereof having the magnetic layer **6** therein faces the substrate **8**. At least part of an end portion of the vapor deposition mask **100** is held by a mask holder **13** made of a magnetic material. The magnet **10** attracts the mask holder **13** and the magnetic layer **6** of the vapor deposition mask **100**, so as to bring the low-hardness layer **7** of the vapor deposition mask **100** into contact with the substrate **8**. The vapor deposition mask **100** is aligned such that the openings **4** face respective areas to be subjected to vapor deposition (hereinafter simply referred to as areas) **11** of the substrate **8**. A vapor deposition material flying from a vapor deposition source **12** (not illustrated) disposed below the vapor deposition mask **100** passes through the openings **4** and adheres to the areas **11**.

[0032] Advantageous effects of the present embodiment will now be described in comparison with comparative examples illustrated in FIG. **3**A and FIG. **3**B.

[0033] A main function of the magnetic layer **6** according to the present embodiment is to hold the vapor deposition mask **100** as parallel as possible to the substrate **8**. For comparison, FIG. **3**A illustrates an example of how the vapor deposition mask **100** and the substrate **8** face each other in the absence of the magnetic layer **6** in the present embodiment. Note that the illustration of the openings **4** and the areas **11** is omitted in FIG. **3**A. In the example illustrated in FIG. **3**A, the vapor deposition mask **100** warps due to its own weight or stress, and cannot be held parallel to the substrate **8**. In the present embodiment, on the other hand, the beam region **3** disposed inside and on the outer periphery of the vapor deposition mask **100** is provided with the magnetic layer **6**. Therefore, even when the vapor deposition mask **100** warps as illustrated in FIG. **3**A, the magnet **10** can attract the magnetic layer **6**, so that the vapor deposition mask **100** can be disposed parallel to the substrate **8**.

[0034] To bring the low-hardness layer **7** of the vapor deposition mask **100** into contact with the substrate **8**, a magnetic force exceeding the weight of the vapor deposition mask **100** may be generated between the magnetic layer **6** and the magnet **10**. The volume of the magnetic layer **6** may be increased to generate a large magnetic force. In the present embodiment, however, the thickness of the magnetic layer **6** is set thicker than the membrane regions **2** and the recess **5**, so that a sufficient volume can be secured.

[0035] In the present embodiment, the base member **1** is made of a non-magnetic material and the membrane regions **2** do not include the magnetic layer **6**. This is to prevent plastic deformation of or damage to the membrane regions **2** when the magnet **10** attracts the vapor deposition mask **100**. For comparison, FIG. **3**B illustrates an example of how the vapor deposition mask **100** and the substrate **8** face each other when the membrane regions **2** are made of a magnetic material. Note that the illustration of the electrostatic chuck **9**, the magnet **10**, and the mask holder **13** is omitted in FIG. **3**B. The membrane regions **2** are generally very thin and easily deformed when subjected to external force. That is, when the magnet **10** attracts the vapor deposition mask **100**, the membrane regions **2** each significantly bulge in the center thereof toward the substrate **8** in the example illustrated in FIG. **3**B and are prone to plastic deformation or damage. In the present embodiment, on the other hand, the membrane regions **2** do not contain a magnetic material and are made only of a non-magnetic material. Therefore, the membrane regions **2** are not deformed by magnetic force and the risk of plastic deformation or damage can be reduced.

[0036] In the present embodiment, the magnetic layer **6** and the low-hardness layer **7** also serve as a spacer, so that the membrane regions **2** are not in contact with the substrate **8**. This can further reduce the risk of damage to the membrane regions **2** caused by contact with the substrate **8**.

[0037] Each component of the vapor deposition mask **100** according to the present embodiment will now be described.

Membrane Region (First Region) **2**

[0038] The membrane regions **2** can be held as flat as possible to the substrate **8**. For this, the membrane regions **2** can be made of a non-magnetic material with a volume magnetic susceptibility of 1 or less, so as to prevent deformation caused by the magnetic field of the magnet **10**. A material selected to form the membrane regions **2** can be a high rigidity material that is resistant to deformation and has a Young's modulus of greater than or equal to 50 GPa, preferably greater than 100 GPa. When the membrane regions **2** have a film structure that exerts tensile stress as a whole, it is easier to maintain a flat state of the membrane regions **2**. For the openings **4** to be arranged with high accuracy and precision, it is even better to select the material for the membrane regions **2** by also taking into consideration the ease of processing. The membrane regions **2** may be formed by using a base material, such as a semiconductor substrate, as it is, or by using chemical vapor deposition (CVD), sputtering, or any of various plating methods.

[0039] The Young's modulus of a main layer of each membrane region **2** is preferably greater than or equal to 50 GPa, more preferably greater than or equal to 100 GPa, and still more preferably greater than 100 GPa. The main layer of the membrane region **2** can be a non-magnetic metal layer. The main layer of the membrane region **2** can contain silicon (Si). Note that the main layer of the membrane region **2** here is the thickest layer of all layers constituting the membrane regions **2**. When there are a plurality of layers made of the same material, the total thickness of the layers made of the same material is the thickness of the layer made of that material. Here, a main layer of a member can be a layer of which thickness is the greatest among layers included in the member.

Beam Region (Second Region) **3**

[0040] The beam region **3** is a component that determines the rigidity and mass of the entire vapor deposition mask **100**. Therefore, the beam region **3** can be made of a material that has not only a rigidity same as or similar to that of the membrane regions **2**, but also a specific gravity lighter than that of the membrane regions **2**. Further, if a material having a small coefficient of linear expansion or a material having a coefficient of linear expansion close to that of the substrate **8** to be adhered is used, alignment deviation due to heat generation between the mask **100** and the substrate **8** to be adhered can be suppressed. The beam region **3** may be formed by using a base material, such as a semiconductor substrate, as it is, or by using CVD, sputtering, or any of various plating methods.

Magnetic Layer **6**

[0041] The magnetic layer **6** can be made of a material that is easily attracted to the magnet **10**. A material selected to form the magnetic layer **6** thus preferably has a volume magnetic susceptibility of greater than or equal to 10, and more preferably has a volume magnetic susceptibility of greater than or equal to 100. Examples of the selectable material include iron, nickel, cobalt, and alloys of these metals. The magnetic layer **6** may be formed by electroplating or electroless plating, or by any of various printing methods, such as screen printing or patterning with a dispenser using an ink made of a magnetic material.

Low-Hardness Layer **7**

[0042] The low-hardness layer **7** is a component that is in direct contact with the substrate **8**. Therefore, the low-hardness layer **7** is required not to damage the substrate **8**. A material selected to form the low-hardness layer **7** can thus be a material having a lower hardness than those forming the base member **1** and the magnetic layer **6**. The low-hardness layer **7** can have a lower hardness than the surface of the magnetic layer **6** and the surface of the membrane regions **2** on the same side as the surface of the beam region **3** having the magnetic layer **6** therein. The low-hardness layer **7** can be made of an organic material, or can contain fluororesin. The low-hardness layer **7** may be formed by using any of various vapor deposition techniques, CVD, sputtering, or any of various plating methods.

Method for Manufacturing Vapor Deposition Mask

[0043] An example of a method for manufacturing the vapor deposition mask according to the present embodiment will now be described using FIG. **4**A to FIG. **4**D and FIG. **5**A to FIG. **5**C. Note that FIG. **4**A to FIG. **4**D illustrate part IV of FIG. **5**A upside down. In FIG. **5**A to FIG. **5**C, the illustration of a stress adjusting layer **17** and the low-hardness layer **7** is omitted.

[0044] A silicon on insulator (SOI) substrate **20** is used as a base material. In the present example, a device layer **14** of the SOI substrate **20** is a main layer of each membrane region **2**. Therefore, the thickness of the device layer **14** is set thicker than a buried oxide (BOX) layer **15**. The device layer **14** is a layer of single crystal silicon, which is high in Young's modulus, light in specific gravity, finely patterned easily, and thus is suitable as a material for forming the membrane region **2**.

[0045] First, as illustrated in FIG. **4**A, the stress adjusting layer **17**, such as a SiN film, is formed on the surface of the device layer **14** of the SOI substrate **20** by plasma-enhanced CVD or the like to a thickness thinner than the device layer **14**. The stress adjusting layer **17** is a surface layer of the membrane region **2** on the same side as the surface of the beam region **3** having the magnetic layer **6** therein. Under tensile stress applied by the stress adjusting layer **17**, the membrane region **2** to be formed later can be easily maintained in a flat state.

[0046] Next, as illustrated in FIG. **4**B, the recess **5** and regions to be turned into the openings **4** are patterned by photolithography. In the present example, not only the stress adjusting layer **17** and the device layer **14**, but also the BOX layer **15** and part of a handle layer **16**, are etched, for example, by reactive ion etching (RIE) using a reactive gas.

[0047] After forming of a seed layer (not illustrated) in the recess **5**, the magnetic layer **6** made of magnetic metal, such as nickel, is formed, as illustrated in FIG. **4**C, to a thickness greater than the depth of the recess **5** by electroplating or the like. Then, for example, electroless nickel and polytetrafluoroethylene (PTFE) composite plating is performed on the magnetic layer **6** to form the low-hardness layer **7**.

[0048] Next, as illustrated in FIG. **4**D, a resist is, for example, air-sprayed over the entire surface of the SOI substrate **20** on the side of the device layer **14** to form a protective layer **18**. Then, a conductive tape **19** is attached onto the top of the protective layer **18** for more protection.

[0049] Next, the SOI substrate **20** to which the conductive tape **19** has been applied is subjected to processing on the side of the handle layer **16**. First, as illustrated in FIG. **5**A, a pattern of the resist

21 is formed, in a region to be turned into the beam region **3**, by photolithography on the surface of the handle layer **16**. Note that openings in the resist **21** become the membrane regions **2**.
[0050] Next, as illustrated in FIG. **5**B, the handle layer **16** is etched to the BOX layer **15**. For example, the Bosch process can be used for etching.
[0051] Next, as illustrated in FIG. **5**C, the conductive tape **19** is removed from the SOI substrate **20**, and the resist **21** and the protective layer **18** are peeled off with an organic solvent to obtain the vapor deposition mask **100**.
[0052] The vapor deposition mask of the first embodiment obtained as described above is a fine mask, as it is processed using a silicon process. Also, the vapor deposition mask of the first embodiment can be reliably attracted to the magnet and is resistant to plastic deformation and damage when attracted to the magnet or when brought into contact with the substrate to be subjected to vapor deposition. It is thus possible to achieve high precision and improved reliability of an organic electronic device, such as an organic light-emitting device.
[0053] A vapor deposition mask can thus be provided, which is resistant to plastic deformation and damage and can be reliably attracted by magnetic force.
Second Embodiment
Vapor Deposition Mask
[0054] FIG. **6** is a schematic cross-sectional view illustrating how vapor deposition is performed using a vapor deposition mask according to a second embodiment. In the present embodiment, the thickness of the magnetic layer **6** embedded in the recess **5** is less than the depth of the recess **5**. The low-hardness layer **7** is absent in the present embodiment. The description of other parts will be omitted, as they are the same as or similar to the first embodiment.
[0055] The thickness of the magnetic layer **6** is less than the depth of the recess **5**, but is large enough because the depth of the recess **5** is substantially greater than the thickness of the membrane region **2**. Therefore, the magnetic layer **6** has a volume large enough for the magnetic layer **6** to be attracted to the magnet **10**.
[0056] In the present embodiment, the magnetic layer **6** does not serve as a spacer between the vapor deposition mask **100** and the substrate **8**. However, such a system is useful particularly when the substrate **8** has spacers **22** thereon, as illustrated in FIG. **6**. The spacers **22** can be formed on the substrate **8** with high accuracy and precision by combining spin coating or CVD with a photolithography process. With the spacers **22**, a gap between the membrane regions **2** and the substrate **8** can be narrowed and defined with high precision, so that the thickness distribution of a film to be vapor-deposited in the areas **11** can be made uniform. In such a system, the thickness of the magnetic layer **6** can be made thinner than the recess **5**, so as to allow the spacers **22** to define the gap with high accuracy.
Method for Manufacturing Vapor Deposition Mask
[0057] An example of a method for manufacturing the vapor deposition mask according to the present embodiment will now be described using FIG. **7**A to FIG. **7**D.
[0058] A silicon wafer **23** is used as a base material. First, as illustrated in FIG. **7**A, a silicon oxide film **24**, a silicon nitride film **25**, and a silicon oxide film **26** are formed in sequence on the surface of the silicon wafer **23**. For example, the silicon oxide films **24** and **26** can be formed by plasma-enhanced CVD, and the silicon nitride film **25** can be formed by low-pressure (LP)-CVD. In the present example, the silicon nitride film **25** becomes a main layer of the membrane region **2**. Therefore, the thickness of the silicon nitride film **25** is set thicker than the total thickness of the silicon oxide films **24** and **26**.
[0059] Next, as illustrated in FIG. **7**B, the recess **5** and regions to be turned into the openings **4** are patterned by photolithography. The silicon oxide films **24** and **26** and the silicon nitride film **25** can be etched, for example, by reactive ion etching (RIE) using a reactive gas. Then, the silicon wafer **23** at the bottom of the openings **4** and the recess **5** is etched, for example, by the Bosch process.
[0060] After forming of a seed layer (not illustrated) in the recess **5**, the magnetic layer **6** made of

magnetic metal, such as nickel, is formed, as illustrated in FIG. **7**C, to a thickness less than the depth of the recess **5** by electroplating or the like.

[0061] Then, the back side of the silicon wafer **23** is processed in a procedure similar to that illustrated in FIG. **4**D to FIG. **5**C of the first embodiment, so that the vapor deposition mask **100** illustrated in FIG. **7**D can be obtained.

[0062] The vapor deposition mask of the second embodiment obtained as described above is a high definition mask, as it is processed using a silicon process. Also, the vapor deposition mask of the second embodiment can be reliably attracted to the magnet and is resistant to plastic deformation and damage when attracted to the magnet or when brought into contact with the substrate to be subjected to vapor deposition. It is thus possible to achieve high precision and improved reliability of an organic electronic device, such as an organic light-emitting device.

Third Embodiment

Vapor Deposition Mask

[0063] FIG. **8** is a schematic cross-sectional view illustrating how vapor deposition is performed using a vapor deposition mask according to a third embodiment. In the present embodiment, the depth of the recess **5** is equal to the thickness of the membrane regions **2**. The description of other parts will be omitted, as they are the same as or similar to the first embodiment.

[0064] In the present embodiment, the magnetic layer **6** is thicker than the membrane regions **2**. Therefore, the magnetic layer **6** has a volume large enough for the magnetic layer **6** to be attracted to the magnet **10**.

Method for Manufacturing Vapor Deposition Mask

[0065] An example of a method for manufacturing the vapor deposition mask according to the present embodiment will now be described using FIG. **9**A to FIG. **9**D.

[0066] The silicon wafer **23** is used as a base material. First, as illustrated in FIG. **9**A, a non-magnetic metal layer **28** is patterned on the silicon wafer **23**. In the present example, an electroless nickel-phosphorus coating is patterned using a semi-additive process. Specifically, first, a palladium (Pd) thin film or the like is formed as a seed layer **27** on the surface of the silicon wafer **23**. Then, a resist pattern (not illustrated) is formed in regions to be turned into the openings **4** and the recess **5**, and subjected to electroless nickel-phosphorus plating. Here, the concentration of phosphorus in nickel is adjusted such that a film obtained by plating is a non-magnetic film. Then, the resist pattern (not illustrated) is removed to obtain the non-magnetic metal layer **28** having the pattern of the recess **5** and the regions to be turned into the openings **4**. In the present example, the non-magnetic metal layer **28** is a main layer of the membrane region **2** and serves as a surface layer of the membrane region **2** on the same side as the surface of the beam region **3** having the magnetic layer **6** therein.

[0067] Next, as illustrated in FIG. **9**B, the seed layer **27** in the regions to be turned into the openings **4** and at the bottom of the recess **5** is removed by wet etching or the like. Then, a magnetic metal-containing ink, such as a magnetite fine-particle ink, is applied by screen printing or the like to the recess **5**, so as to pattern the magnetic layer **6** thicker than the membrane region **2**.

[0068] Next, as illustrated in FIG. **9**C, a resin, such as an acrylic resist (acrylic resin), is applied with a dispenser or the like to the magnetic layer **6** to form the low-hardness layer **7**.

[0069] Then, the back side of the silicon wafer **23** is processed in a procedure similar to that illustrated in FIG. **4**D to FIG. **5**C of the first embodiment, so that the vapor deposition mask **100** illustrated in FIG. **9**D can be obtained.

[0070] The vapor deposition mask of the third embodiment obtained as described above can be manufactured at lower cost, as it does not require excessive use of a vacuum process. Also, the vapor deposition mask of the third embodiment can be reliably attracted to the magnet and is resistant to plastic deformation and damage when attracted to the magnet or when brought into contact with the substrate to be subjected to vapor deposition. It is thus possible to achieve high precision and improved reliability of an organic electronic device, such as an organic light-emitting device.

Manufacturing Method and Application of Organic Electronic Device

[0071] A method for manufacturing an organic electronic device according to the present embodiment includes, as illustrated in FIG. **2**, placing the vapor deposition mask **100** of the present embodiment, with the surface thereof having the magnetic layer **6** therein facing the substrate **8**; and vapor-depositing an organic material on the substrate **8**. Examples of the organic electronic device according to the present embodiment include an organic light-emitting device, such as an organic EL display device.

[0072] An organic light-emitting device according to the present embodiment can be used as a component of a display apparatus or an illuminating apparatus. Other applications of the organic light-emitting device according to the present embodiment include an exposing source for an electrophotographic image forming apparatus, a backlight for a liquid crystal display apparatus, or a light-emitting device including a color filter and a white light source.

[0073] The display apparatus may be an image information processing apparatus that includes an image input unit configured to receive image information from an area charge-coupled device (CCD), a linear CCD, a memory card, or the like, and an information processing unit configured to process the received information, and is configured to display the received image on a display unit. The display apparatus may include the organic light-emitting device of the present embodiment. The organic light-emitting device may include a plurality of pixels, and at least one of the plurality of pixels may include an organic light-emitting element and an active element, such as a transistor, connected to the organic light-emitting element. Here, the substrate may be a semiconductor substrate, such as a silicon substrate, and the transistor may be a metal-oxide-semiconductor field-effect transistor (MOSFET) on the substrate. An image display apparatus includes an input unit configured to receive image information and a display unit configured to output an image, and the display unit includes a display device of the present embodiment.

[0074] A display unit included in an imaging device or an inkjet printer may have a touch panel function. A method for driving the touch panel function is not particularly limited, and may be of an infrared type, a capacitive type, a resistive film type, or an electromagnetic induction type. The display device may be used as a display unit of a multifunction printer.

[0075] FIG. **10** schematically illustrates an example of the display device according to the present embodiment. A display device **1000** may include, between an upper cover **1001** and a lower cover **1009**, a touch panel **1003**, a display panel **1005**, a frame **1006**, a circuit board **1007**, and a battery **1008**. Flexible printed circuits (FPCs) **1002** and **1004** are connected to the touch panel **1003** and the display panel **1005**, respectively. A transistor is printed on the circuit board **1007**. The battery **1008** is optional when the display device is not a mobile device. Even when the display device is a mobile device, the battery **1008** may be provided in a different place.

[0076] The display device according to the present embodiment may include a color filter of red, green, and blue. The red, green, and blue of the color filter may be arranged in a delta configuration.

[0077] The display device according to the present embodiment may be used as a display unit of a mobile terminal. In this case, the display device may have both a display function and an operation function. Examples of the mobile terminal include mobile phones, such as smartphones, tablets, and head-mounted displays (HMDs).

[0078] The display device according to the present embodiment may be used as a display unit of an imaging device that includes an optical unit including a plurality of lenses and an imaging element configured to receive light passing through the optical unit. The imaging device may include a display unit configured to display information acquired by the imaging element. The display unit may be either one that is exposed to the outside of the imaging device, or one that is disposed inside the finder. The imaging device may be a digital camera or a digital video camera.

[0079] FIG. **11**A schematically illustrates an example of an imaging device according to the present embodiment. An imaging device **1100** may include a viewfinder **1101**, a back display **1102**, an operation unit **1103**, and a housing **1104**. The viewfinder **1101** may include the display device according to the present embodiment. In this case, the display device may display not only an image to be captured, but also environmental information, imaging instructions, and the like. The environmental information may include the intensity of external light, the direction of external light, the moving speed of the subject, and the possibility that the subject will be blocked by an obstruction.

[0080] Since the timing suitable for capturing an image is limited, it is better to display information as quickly as possible. Therefore, a display apparatus using the organic light-emitting device of the present embodiment can be used. This is because the organic light-emitting device has a high response speed. The display apparatus using the organic light-emitting device can be used more preferably than a liquid crystal display apparatus that is required to have a high display speed.

[0081] The imaging device **1100** includes an optical unit (not illustrated). The optical unit includes a plurality of lenses and is configured to form an image on an imaging element housed in the housing **1104**. Focus can be adjusted by adjusting the relative position of the plurality of lenses. This operation may be done automatically. The imaging device may also be referred to as a photoelectric conversion apparatus. Instead of sequentially capturing images, the photoelectric conversion apparatus may detect a difference from the preceding image, or may cut out part of an image constantly recorded, to capture an image.

[0082] FIG. **11**B schematically illustrates an example of an electronic apparatus according to the present embodiment. An electronic apparatus **1200** includes a display unit **1201**, an operation unit **1202**, and a housing **1203**. The housing **1203** may include a circuit, a printed board including the circuit, a battery, and a communicating unit. The display unit **1201** may include the organic light-emitting device according to the present embodiment. The operation unit **1202** may be a button or a reactive unit of touch panel type. The operation unit **1202** may be a biometric recognition unit configured to recognize a fingerprint for unlocking or the like. The electronic apparatus including the communicating unit may also be referred to as a communication apparatus. The electronic apparatus **1200** may include a lens and an imaging element to further provide a camera function. An image captured by the camera function is displayed on the display unit **1201**. Examples of the electronic apparatus **1200** include smartphones and laptop computers.

[0083] FIG. **12**A and FIG. **12**B each schematically illustrate an example of the display device according to the present embodiment. FIG. **12**A illustrates a display device, such as a TV monitor or a PC monitor. A display device **1300** includes a frame **1301** and a display unit **1302**. The display unit **1302** may include a light-emitting device according to the present embodiment. The display device **1300** includes a base **1303** configured to support the frame **1301** and the display unit **1302**. The base **1303** is not limited to that illustrated in FIG. **12**A. The lower side of the frame **1301** may also serve as a base. The frame **1301** and the display unit **1302** may be curved, and their radius of curvature may be greater than or equal to 5000 mm and less than or equal to 6000 mm.

[0084] FIG. **12**B schematically illustrates another example of the display device according to the present embodiment. A display device **1310** illustrated in FIG. **12**B is a so-called foldable display device configured to be capable of being folded up. The display device **1310** includes a first display unit **1311**, a second display unit **1312**, a housing **1313**, and a folding point **1314**. The first display unit **1311** and the second display unit **1312** may include the light-emitting device according to the present embodiment. The first display unit **1311** and the second display unit **1312** may be a single seamless display device. The first display unit **1311** and the second display unit **1312** may be separated at the folding point **1314**. The first display unit **1311** and the second display unit **1312** may each display a different image, or may display a single image together.

[0085] FIG. **13**A schematically illustrates an example of an illuminating apparatus according to the present embodiment. An illuminating apparatus **1400** may include a housing **1401**, a light source **1402**, a circuit board **1403**, an optical filter **1404** configured to transmit light emitted by the light source **1402**, and a light diffuser **1405**. The light source **1402** may include the organic light-emitting device according to the present embodiment. The optical filter **1404** may be a filter that improves color rendering properties of the light source **1402**. The light diffuser **1405** can effectively diffuse light from the light source **1402** and distribute the light over a wide area for lighting up or the like. The optical filter **1404** and the light diffuser **1405** may be disposed on the light emerging side of the illumination, and may be provided with a cover on the outermost side as necessary.

[0086] The illuminating apparatus is, for example, an apparatus configured to illuminate a room. The illuminating apparatus may emit light of white, neutral white, or any other color selected from the range of blue to red colors. The illuminating apparatus may include a dimmer circuit configured to control the light or a toning circuit configured to control the color of emitted light. The illuminating apparatus may include the organic light-emitting device of the present embodiment and a power supply circuit connected thereto. The power supply circuit is a circuit configured to convert an alternating-current (AC) voltage into a direct-current (DC) voltage. The illuminating apparatus may include an inverter circuit. White has a color temperature of 4200 K, and neutral white has a color temperature of 5000 K. The illuminating apparatus may include a color filter.

[0087] The illuminating apparatus according to the present embodiment may include a heat dissipating unit. The heat dissipating unit is configured to release heat from inside to outside of the apparatus. For example, the heat dissipating unit may be made of metal with high specific heat, or liquid silicon.

[0088] FIG. **13**B schematically illustrates a car that is an example of a moving body according to the present embodiment. The car includes a tail lamp that is an example of a lighting tool. A car **1500** may include tail lamps **1501** and may be configured to turn on the tail lamps **1501** when brakes are applied.

[0089] The tail lamps **1501** may each include the organic light-emitting device according to the present embodiment. The tail lamps **1501** may each include a protective member configured to protect the organic light-emitting device. The protective member may be made of any material that is transparent and has a certain degree of strength, but can be made of polycarbonate or the like. For example, the polycarbonate may be mixed with a furandicarboxylic acid derivative or an acrylonitrile derivative.

[0090] The car **1500** may include a car body **1503** and a window **1502** attached thereto. The window **1502** may be a transparent display unless it is intended for viewing the front or rear of the car **1500**. The transparent display may include the organic light-emitting device according to the present embodiment. In this case, components of the organic light-emitting device, such as electrodes, are transparent.

[0091] The moving body according to the present embodiment may be, for example, a ship, an aircraft, or a drone. The moving body may include an airframe and a lighting tool included in the airframe. The lighting tool may emit light to indicate the location of the airframe. The lighting tool includes the organic light-emitting device according to the present embodiment.

[0092] With reference to FIG. **14**A and FIG. **14**B, applications of the display device according to each of the aforementioned embodiments will now be described. The display device is applicable to systems that can be worn as wearable devices, such as smart glasses, HMDs, and smart contact lenses. An imaging and display apparatus used in such applications includes an imaging device capable of photoelectrically converting visible light and a display device capable of emitting visible light.

[0093] FIG. **14**A schematically illustrates an example of a wearable device according to an embodiment of the present disclosure. Eyeglasses **1600** (smart glasses) according to an application will now be described using FIG. **14**A. An imaging device **1602**, such as a complementary metal-oxide-semiconductor (CMOS) sensor or a single-photon avalanche diode (SPAD), is disposed on the surface of a lens **1601** of the eyeglasses **1600**. The display device according to any of the aforementioned embodiments is disposed on the back side of the lens **1601**.

[0094] The eyeglasses **1600** further include a controller **1603**. The controller **1603** serves as a power source configured to supply power to the imaging device **1602** and the display device. The controller **1603** is also configured to control the operation of the imaging device **1602** and the display device. The lens **1601** has an optical system for focusing light on the imaging device **1602**.

[0095] FIG. **14**B schematically illustrates another example of the wearable device according to the embodiment of the present disclosure. Eyeglasses **1610** (smart glasses) according to an application will now be described using FIG. **14**B. The eyeglasses **1610** include a controller **1612**, which includes an imaging device corresponding to the imaging device **1602** illustrated in FIG. **14**A and a display device. A lens **1611** has an optical system for projecting light emitted from the imaging device and the display device included in the controller **1612**, and an image is projected onto the lens **1611**. The controller **1612** not only serves as a power source configured to supply power to the imaging device and the display device, but also controls the operation of the imaging device and the display device.

[0096] The controller **1612** may include a line-of-sight detector configured to detect a wearer's line of sight. The line of sight may be detected using infrared light. An infrared light-emitting unit emits infrared light toward an eyeball of the user who is gazing at a displayed image. The infrared light emitted is reflected from the eyeball and detected by an imaging unit including a light receiving element, so that a captured image of the eyeball is obtained. With a reducing unit configured to reduce light from the infrared light-emitting unit to a display unit in plan view, a degradation of image quality is reduced. From the captured image of the eyeball obtained by infrared imaging, a user's line of sight toward the displayed image is detected. Any known technique can be applied to line-of-sight detection using the captured image of the eyeball. For example, a line-of-sight detecting method based on a Purkinje image formed by reflection of irradiation light on the cornea can be used. More specifically, line-of-sight detection processing based on a pupil-corneal reflection method is performed. The user's line of sight is detected by using the pupil-corneal reflection method and calculating a line-of-sight vector representing the orientation (rotation angle) of the eyeball, on the basis of a pupil image included in the captured image of the eyeball and the Purkinje image.

[0097] A display device according to an embodiment of the present disclosure may include an imaging device including a light receiving element, and may control an image displayed on the display device on the basis of user's line-of-sight information from the imaging device. Specifically, on the basis of the line-of-sight information, the display device determines a first field-of-view region that the user is gazing at and a second field-of-view region outside the first field-of-view region. The first field-of-view region and the second field-of-view region may be either determined by a controller of the display device, or may be determined by and received from an external controller. In a display region of the display device, a display resolution in the first field-of-view region may be controlled to be higher than a display resolution in the second field-of-view region. That is, the resolution in the second field-of-view region may be lower than that in the first field-of-view region.

[0098] The display region includes a first display region and a second display region different from the first display region, and a high-priority region is determined from the first display region and the second display region on the basis of line-of-sight information. The first field-of-view region and the second field-of-view region may be either determined by the controller of the display device, or may be determined by and received from an external controller. A resolution in the high-priority region may be controlled to be higher than a resolution in a region other than the high-priority region. That is, a resolution in a lower-priority region may be set to be low.

[0099] The first field-of-view region and the high-priority region may be determined using artificial intelligence (AI). The AI may be a model configured to estimate, from an image of an eyeball, the angle of line of sight and the distance to an object at which the line of sight is directed, by using the image of the eyeball and the direction in which the eyeball in the image is actually looking as teaching data. The AI program may be included in the display device, the imaging device, or an external device. If included in the external device, the AI program is transmitted via communication to the display device.

[0100] For display control based on visual recognition detection, the display device is applicable to smart glasses that further include an imaging device configured to capture images of the outside. The smart glasses are capable of displaying captured outside information in real time.

[0101] FIG. **15**A schematically illustrates an example of an image forming apparatus according to an embodiment of the present disclosure. An image forming apparatus **140** is an electrophotographic image forming apparatus and includes a photosensitive member **127**, an exposing source **128**, a charging unit **130**, a developing unit **131**, a transfer unit **132**, conveying rollers **133**, and a fuser **135**. Light **129** is emitted from the exposing source **128** to form an electrostatic latent image on the surface of the photosensitive member **127**. The exposing source **128** includes the organic light-emitting device according to the present embodiment. The developing unit **131** includes toner or the like. The charging unit **130** is configured to charge the photosensitive member **127**. The transfer unit **132** is configured to transfer a developed image onto a recording medium **134**. The conveying rollers **133** are configured to convey the recording medium **134**. For example, the recording medium **134** is paper. The fuser **135** is configured to fix an image formed on the recording medium **134**.

[0102] FIG. **15**B and FIG. **15**C are diagrams each illustrating the exposing source **128**. FIG. **15**B and FIG. **15**C each schematically illustrate how a plurality of light-emitting portions **136** are arranged on a long substrate. An arrow **137** represents a direction parallel to the axis of the photosensitive member **127** and indicates a column direction in which organic light-emitting elements are arranged. The column direction is the same as the direction of the axis on which the photosensitive member **127** rotates, and can also be referred to as a longitudinal direction of the photosensitive member **127**. FIG. **15**B illustrates the light-emitting portions **136** arranged along the longitudinal direction of the photosensitive member **127**. Unlike the light-emitting portions **136** illustrated in FIG. **15**B, the light-emitting portions **136** illustrated in FIG. **15**C are alternately arranged in the column direction, in each of a first column and a second column. The first column and the second column are at different positions in the row direction. In the first column, the plurality of light-emitting portions **136** are arranged at intervals. In the second column, the light-emitting portions **136** are arranged at positions corresponding to spaces between adjacent ones of the light-emitting portions **136** in the first column. That is, the plurality of light-emitting portions **136** are arranged at intervals also in the row direction. The arrangement illustrated in FIG. **15**C may also be referred to as a grid pattern, a staggered arrangement, or a checkered pattern.

[0103] As described above, with the organic light-emitting device according to the present embodiment, it is possible to provide stable display of good image quality even for a long time. Also, with the organic light-emitting device according to the present embodiment, it is possible to achieve both good visibility outdoors and power-saving display because of highly efficient and high-brightness light output.

EXAMPLES

Example 1

[0104] In the present example, the vapor deposition mask illustrated in FIG. **1**A and FIG. **1**B was manufactured in the procedure illustrated in FIG. **4**A to FIG. **4**D and FIG. **5**A to FIG. **5**C.

[0105] The SOI substrate **20** was used as a base material. In the present example, the device layer **14** of the SOI substrate **20** is a main material for forming each membrane region **2**. Therefore, the thickness of the device layer **14** is set thicker than the BOX layer **15**. The device layer **14** is a layer of single crystal silicon, which is high in Young's modulus, light in specific gravity, finely patterned easily, and thus is suitable as a material for forming the membrane region **2**. The Young's modulus and the indentation hardness of the device layer **14**, measured with a nanoindentation technique, were 135 GPa and 11.3 GPa, respectively.

[0106] First, as illustrated in FIG. **4**A, a SiN film was formed as the stress adjusting layer **17** on the surface of the device layer **14** of the SOI substrate **20** by plasma-enhanced CVD to a thickness thinner than the device layer **14**. The stress adjusting layer **17** is a surface layer of the membrane region **2** on the same side as the surface of the beam region **3** having the magnetic layer **6** therein. Under tensile stress applied by the stress adjusting layer **17**, the membrane region **2** to be formed later can be easily maintained in a flat state. The membrane stress of the SiN film was a tensile stress of 150 MPa. The Young's modulus and the indentation hardness of the SiN film were 140 GPa and 7.5 GPa, respectively.

[0107] Next, as illustrated in FIG. **4**B, the recess **5** and regions to be turned into the openings **4** were patterned by photolithography. Not only the stress adjusting layer **17** and the device layer **14**, but also the BOX layer **15** and part of the handle layer **16**, were etched by reactive ion etching (RIE) using a reactive gas.

[0108] After forming a seed layer (not illustrated) in the recess **5**, the magnetic layer **6** made of nickel was formed, as illustrated in FIG. **4**C, to a thickness 1.0 μm greater than the depth of the recess **5** by electroplating. Here, the indentation hardness of the magnetic layer **6** was 6.1 GPa. Then, electroless nickel and PTFE composite plating was performed on the magnetic layer **6** to form the low-hardness layer **7**. The thickness and the indentation hardness of the low-hardness layer **7** were 1.0 μm and 2.6 GPa, respectively.

[0109] Next, as illustrated in FIG. **4**D, a resist was air-sprayed over the entire surface of the SOI substrate **20** on the side of the device layer **14** to form the protective layer **18**. Then, the conductive tape **19** was attached to the top of the protective layer **18** for more protection.

[0110] Next, the SOI substrate **20** to which the conductive tape **19** had been attached was subjected to processing on the side of the handle layer **16**. First, as illustrated in FIG. **5**A, a pattern of the resist **21** was formed, in a region to be turned into the beam region **3**, by photolithography on the surface of the handle layer **16**. Note that openings in the resist **21** become the membrane regions **2**.

[0111] Next, as illustrated in FIG. **5**B, the handle layer **16** was etched to the BOX layer **15**. A Bosch process was used for etching.

[0112] Next, as illustrated in FIG. **5**C, the conductive tape **19** was removed from the SOI substrate **20**, and the resist **21** and the protective layer **18** were peeled off with an organic solvent to obtain the vapor deposition mask **100**.

Example 2

[0113] In the present example, the vapor deposition mask illustrated in FIG. **6** was manufactured in the procedure illustrated in FIG. **7**A to FIG. **7**D.

[0114] The silicon wafer **23** was used as a base material. First, as illustrated in FIG. **7**A, the silicon oxide film **24**, the silicon nitride film **25**, and the silicon oxide film **26** were formed in sequence on the surface of the silicon wafer **23**. The silicon oxide films **24** and **26** were formed by plasma-enhanced CVD, and the silicon nitride film **25** was formed by LP-CVD. In the present example, the silicon nitride film **25** becomes a main layer of the membrane region **2**. Therefore, the thickness of the silicon nitride film **25** was set thicker than the total thickness of the silicon oxide films **24** and **26**. The membrane stress of the silicon nitride film **25** was a tensile stress of 160 MPa. The Young's modulus and the indentation hardness of the silicon nitride film **25** were 230 GPa and 7.8 GPa, respectively.

[0115] Next, as illustrated in FIG. **7**B, the recess **5** and regions to be turned into the openings **4** were patterned by photolithography. The silicon oxide films **24** and **26** and the silicon nitride film **25** were etched by reactive ion etching (RIE) using a reactive gas. Then, the silicon wafer **23** at the bottom of the openings **4** and the recess **5** was etched by the Bosch process.

[0116] After forming of a seed layer (not illustrated) in the recess **5**, the magnetic layer **6** made of nickel was formed, as illustrated in FIG. **7**C, to a thickness 0.5 μm less than the depth of the recess **5** by electroplating.

[0117] Then, the back side of the silicon wafer **23** was processed in a procedure similar to that illustrated in FIG. **4**D to FIG. **5**C of Example 1, so that the vapor deposition mask **100** illustrated in FIG. **7**D was obtained.
Example 3
[0118] In the present example, the vapor deposition mask illustrated in FIG. **8** was manufactured in the procedure illustrated in FIG. **9**A to FIG. **9**D.
[0119] The silicon wafer **23** was used as a base material. First, as illustrated in FIG. **9**A, an electroless nickel-phosphorus plating was patterned as the non-magnetic metal layer **28** on the silicon wafer **23** using a semi-additive process. Specifically, first, a Pd thin film was formed as the seed layer **27** on the surface of the silicon wafer **23**. Then, a resist pattern (not illustrated) was formed in regions to be turned into the openings **4** and the recess **5** and subjected to electroless nickel-phosphorus plating. Here, the concentration of phosphorus in nickel was set to 12 wt % such that a film obtained by plating was a non-magnetic film. Then, the resist pattern (not illustrated) was removed to obtain the non-magnetic metal layer **28** having the pattern of the recess **5** and the regions to be turned into the openings **4**. In the present example, the non-magnetic metal layer **28** is a main layer of the membrane region **2** and serves as a surface layer of the membrane region **2** on the same side as the surface of the beam region **3** having the magnetic layer **6** therein. The membrane stress of the non-magnetic metal layer **28** was a tensile stress of 20 MPa. The Young's modulus and the indentation hardness of the non-magnetic metal layer **28** were 50 GPa and 5.5 GPa, respectively.
[0120] Next, as illustrated in FIG. **9**B, the seed layer **27** in the regions to be turned into the openings **4** and at the bottom of the recess **5** was removed by wet etching. Then, a magnetite fine-particle ink was applied by screen printing to the recess **5**, so as to pattern the magnetic layer **6**. The thickness of the magnetic layer **6** was 5.0 μm greater than the thickness of the membrane region **2**. The indentation hardness of the magnetic layer **6** was 4.5 GPa.
[0121] Next, as illustrated in FIG. **9**C, an acrylic resist (acrylic resin) was applied with a dispenser to the magnetic layer **6** to form the low-hardness layer **7**. The thickness and the indentation hardness of the low-hardness layer **7** were 2.0 μm and 1.9 GPa, respectively.
[0122] Then, the back side of the silicon wafer **23** was processed in a procedure similar to that illustrated in FIG. **4**D to FIG. **5**C of Example 1, so that the vapor deposition mask **100** illustrated in FIG. **9**D was obtained.
[0123] The present disclosure can provide a vapor deposition mask that is resistant to plastic deformation and can be reliably attracted by magnetic force. It is thus possible to achieve high precision and improved reliability of an organic electronic device, such as an organic light-emitting device.
[0124] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
[0125] This application claims the benefit of Japanese Patent Application No. 2024-028534 filed Feb. 28, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A vapor deposition mask comprising: a base member made of a non-magnetic material, wherein the base member includes a first region having a plurality of through holes and a second region thicker than the first region; and the second region has a recess including a magnetic layer therein.
2. The vapor deposition mask according to claim 1, wherein the second region is disposed around the first region.
3. The vapor deposition mask according to claim 1, wherein the second region is disposed at an end

portion of the base member.

4. The vapor deposition mask according to claim 1, wherein the recess is in a grid form.

5. The vapor deposition mask according to claim 1, wherein the recess is disposed along an outer periphery of the base member.

6. The vapor deposition mask according to claim 1, wherein the first region is disposed to be flush with a surface of the second region having the magnetic layer therein.

7. The vapor deposition mask according to claim 1, wherein the first region is made only of a non-magnetic material.

8. The vapor deposition mask according to claim 1, wherein a depth of the recess is greater than a thickness of the first region.

9. The vapor deposition mask according to claim 1, wherein a thickness of the magnetic layer is less than a depth of the recess.

10. The vapor deposition mask according to claim 1, wherein a thickness of the magnetic layer is greater than a depth of the recess.

11. The vapor deposition mask according to claim 1, wherein a main layer of the first region has a Young's modulus of greater than or equal to 50 GPa.

12. The vapor deposition mask according to claim 1, wherein a main layer of the first region is a non-magnetic metal layer.

13. The vapor deposition mask according to claim 1, wherein a main layer of the first region contains silicon.

14. The vapor deposition mask according to claim 1, further comprising a low-hardness layer on a surface of the magnetic layer, wherein the low-hardness layer has a lower hardness than a surface of the magnetic layer and a surface of the first region on the same side as a surface of the second region having the magnetic layer therein.

15. The vapor deposition mask according to claim 14, wherein the low-hardness layer is made of an organic material.

16. The vapor deposition mask according to claim 14, wherein the low-hardness layer contains fluororesin.

17. A method for manufacturing an organic electronic device, the method comprising: placing the vapor deposition mask according to claim 1, with a surface thereof having the magnetic layer therein facing a substrate to be subjected to vapor deposition; and vapor-depositing an organic material on the substrate.
